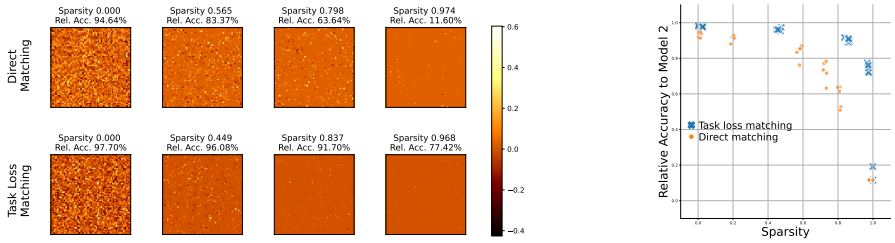

Figure 8: Comparison of stitching matrices and the relative accuracy of L1-regularized direct and task loss matching. To achieve sparsity, we set every entry of the stitching matrix to zero when the absolute value of the entry is below the threshold 10^{-4}. Sparsity is the ratio between the zero elements and the total number of elements in the stitching matrix. Relative accuracies with zero sparsity correspond to the unregularised baseline. **Left:** Examples of stitching matrices with different sparsity. **Right:** The degradation of relative accuracy with respect to increasing sparsity is significantly less steep with task loss training. For more results, see Appendix A.9.

holds on most layers, i.e. the convex combination of the stitching layers perform similarly. However, on the first three layers, we noticed an unforeseen lack of mode connectivity, even though the relative accuracies of the corresponding transformations were high. For further details, see Appendix A.8.

Experiment 7: Sparsity — direct vs. task loss matching How does the sparsity of the stitching layer influence the performance of the stitched network? In this experiment, we achieve sparsity by adding an L1-regularizing term to the training as detailed in Section 4.2, similarly to the experiments in Li et al. [2016]. We perform both an L1-regularized task loss matching as well as a direct loss matching with loss term (3). To obtain a truly sparse stitching matrix we set the elements to zero below a threshold. Sparsity is then computed as the ratio between the zero elements and the total number of elements in the matrix. Figure 8 shows the dependence of relative accuracy on sparsity compared for task loss and direct matching as well as a sample of the stitching matrices. As Figure 8 shows, at around 80-90 $\%$ sparsity, the relative accuracy of direct matching suffers a drastic drop, however task loss matching still manages to find a meaningful alignment of the neurons using additional information via the task loss. These experiments show that even though it is possible to achieve a certain degree of sparse alignment of the neurons, the relationship between two representations is typically not of a combinatorial nature, and much can be gained by considering more complex relationships between the two layers. Furthermore, the difference in the relative accuracy of direct and task loss matching is notable for higher values of sparsity. For further details, see Appendix A.9.

9 Limitations

The performance of the stitched solution is inherently just a lower bound due to possible local minima. The current study is restricted to well-established convolutional architectures. Our observations on network width is analysed only in isolation.

10 Conclusions

We studied similarity of neural network representations in conjunction with the notion of matchability. We demonstrated the matchability of representations in convolutional networks through several experiments and analysed the properties of the stitching transformations with respect to the task performance. Regarding similarity notions, we provided a novel perspective which incorporates the performance on a task; a perspective which is yet unexplored in the field. We pointed out weaknesses of popular similarity notions that end-users might encounter.

Broader impact

Our research is of foundational nature and advances machine learning at a fairly general level. While the authors do not foresee that the work herein will have immediate adverse ethical or societal consequences, such analysis of representation learning could help a malicious user avoid detection that is based on representational similarity; on the other hand, it could move forward the field towards more transparent and interpretable machine learning models.

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